



"Super Duper Secure Mode"

A new mode in the Edge browser makes it harder for attackers to exploit bugs and vulnerabilities. <https://dgit.in/sep21-45>



New Google Nest cams!

Google will allow the new Nest Cams to record local video without a monthly subscription. <https://dgit.in/sep21-46>

GPU Features	GTX 1080Ti	RTX 2080 Ti	Quadro P6000	Quadro RTX 6000
Register File Size/SM	256 KB	256 KB	256 KB	256 KB
Register File Size/GPU	7168 KB	17408 KB	7680 KB	18432 KB
TDP* (Reference/Founders Edition)	250 / 250 W	250 / 260 W	250 W	260 W
Transistor Count	12 Billion	18.6 Billion	12 Billion	18.6 Billion
Die Size	471	754	471	754
Manufacturing Process	16 nm	12 nm FFN	16 nm	12 nm FFN
Note: * Peak TFLOPS, TIPS, and TOPS rates are based on GPU Boost Clock. * Power figure represents Graphics Card TDP only. Note that use of the VirtualLink™/USB Type-C™ connector requires up to an additional 35 W of power that is not represented in this power figure.				

Max-Q and Max-P too, which just adds up to the confusion. So, before splurging on any laptop by just reading the specs, make sure to go through a detailed review from some verified online sources to cross-verify the actual performance of that beautiful beast.

Although TGP is the new buzz word around as RTX devices continue to flow in, it never hurts to know more, so here are some relatively important terms!

THEN WHAT IS TDP?

Every chip holds some extent of thermal threshold limit which actually allows the device to perform properly within the range. The thermal design power (TDP), sometimes also called the thermal design point, is the maximum amount of heat generated by a computer chip or component (often a CPU, GPU or system on a chip) that the cooling system in a computer is designed to dissipate under any workload. For instance, a CPU may have a TDP of 90W, and therefore is expected



to output 90W worth of heat when under heavy use, but it doesn't mean the processor will need to draw out 90W worth of power from the power supply (generally the confusion

is caused since the measurement of TDP is in Watts, the same unit for power measurement). So with higher TDP while you might be able to push your chip to higher performance, it also means you would have to keep an eye on the cooling for that device too. It's essentially a reading that helps determine the 'power efficiency' and performance of a component (NOT the power), and thus is a good thing to keep track of, while shopping for a new laptop/component.

WAIT- WHAT ABOUT TBP?

Well, Total Board Power (or Typical Board Power) is the parameter used by AMD for their Radeon graphics cards and is quite similar to TGP. Although Total Board Power differs from TGP in one very basic aspect which is that it affects the total as well as the peak consumption of the card. Thus, the TBP of a card is a slightly higher number than the TGP of the card and therefore gives a better idea of the peak consumption that the card might reach under load. This is mainly because the TBP number also includes the heat given off by the cooling system of the component, the fans, and sometimes even the lighting system as well. Including all these factors into the total consumption number makes quite the sense as it provides an idea of the peak consumption that the device can potentially reach under certain con-

ditions. With that point off the list, let's proceed further!

WHOA THERE'S STILL MORE? MPC?

NVIDIA is known to use more certain hoopla terms like MPC (or Max Power Consumption), in the specifications of some cards like the Quadro RTX, although the MPC is not equal to the TGP as even NVIDIA claims. In simple words, we can say the MPC is max power consumption supported by the graphics board without damaging it while the TGP/TBP is



SPECIFICATIONS

GPU Memory	48 GB GDDR6
Memory Interface	384-bit
Memory Bandwidth	672 GB/s
ECC	Yes
NVIDIA CUDA Cores	4,608
NVIDIA Tensor Cores	576
NVIDIA RT Cores	72
Single-Precision Performance	16.3 TFLOPS
Tensor Performance	130.5 TFLOPS
NVIDIA NVLink	Connects 2 Quadro RTX 8000 GPUs ¹
NVIDIA NVLink bandwidth	100 GB/s (bidirectional)
System Interface	PCI Express 3.0 x 16
Power Consumption	Total board power: 295 W Total graphics power: 260 W
Thermal Solution	Active
Form Factor	4.4" H x 10.5" L, Dual Slot, Full Height

bearing the max electrical power consumption of the whole graphics card under heavy loads. Thus, it would always be greater than the actual TGP, keep it in mind!

Whew those were quite a few confusing terms. We hope you understood some of it at least and would definitely pay attention when you're all set for your next graphical beast hunt! Do let us know if this helped you understand the terms easier and cleared some misunderstandings, or just left you more flabbergasted. [d](#)